Classical Model of Trade: Ricardian

ECON 404: Cal Poly

Definitions

- **Marginal Product of Labor:** The extra output obtained by using one more unit of labor.
- **Absolute advantage:** The ability of a country to produce a good using fewer productive inputs than is possible anywhere else in the world; or producing the most output with the same amount of inputs.
- **Comparative advantage:** A country has comparative advantage in good x if it gives up less of good y to produce one unit of good x.

Assumptions

1. Perfect Competition in all markets.
2. Factors of production cannot move between countries.
3. There are no barriers to trade in goods.
4. Exports must pay for imports.
5. Labor is the only relevant factor of production in terms of productivity analysis or costs of production.
6. Production exhibits **constant returns to scale** between labor and output.
7. Constant MPL.
8. Foreign variables will be denoted with a '$*$'.

First Theory for Why Countries Trade and Gain from Trade

- Adam Smith, the "first" economist provides an explanation in his 1776 book, *The Wealth of Nations*. Countries can gain by specializing in what they do well and then trading for other goods that they do not produce as easily.
- In fact, we all do this as individuals. I specialize in being a professor and then trade these services to get food, clothing, housing, etc. In contrast, imagine a world where we all had to be self-sufficient.
- Smith laid out a simple narrative to show how this works. The following is a simple numerical model that captures the main elements of his argument.
- Suppose that there are just two countries, England and Portugal, and two goods, Cloth and Wheat. And the following chart shows how much output 1 labor produces of each good in each country.

	Cloth	**Wheat**
England	1 yard	1/4 bushel
Portugal	1/2 yard	1/3 bushel

- **Autarky Scenario:** Assume each country has 24 units of labor (called their *Endowment*). Suppose they put half their time into each good.

  England: Cloth = 12 yards; Wheat = 3 bushels
  Portugal: Cloth = 6 yards; Wheat = 4 bushels
  World: Cloth = 18 yards; Wheat = 7 bushels

	Cloth	**Wheat**
England	1 yard	1/4 bushel
Portugal	1/2 yard	1/3 bushel

- **Specialization and Trade Scenario:** First, let each country specialize in what they do well (i.e., their absolute advantage). Thus, production will be the following:

  England: Cloth = 24 yards; Wheat = 0 bushels
  Portugal: Cloth = 0 yards; Wheat = 8 bushels
  World: Cloth = 24 yards; Wheat = 8 bushels

- What has happened to world production?
  World: Cloth = 18 yards; Wheat = 7 bushels
  World: Cloth = 24 yards; Wheat = 8 bushels
- One possible trading price here that works is 2C for 1W

England: Could trade 6 cloth for 3 Wheat and have 18 yards of cloth and 3 bushels of Wheat.
Portugal: Could trade 3 Wheat for 6 cloth and have 6 yards of cloth and 5 bushels of Wheat.

Take away: If countries specialize in the good they have an *absolute advantage* in and trade, they can consume more than if they stay in autraky.

David Ricardo and Theory of Comparative Advantage

- David Ricardo writes a follow-up piece to Adam smith in 1791, titled *Principles of Political Economy and Taxation*. Smith's example only covers situations where each country has a clear absolute advantage in a different good.
- Ricardo shows that gains from specialization and trade can occur even when one country is more efficient at producing both goods. The reason comes down to opportunity costs.
- To see this we will work through a model in more detail.

Home Country: Supply Side

- Two goods: Cloth (C) and Wheat (W)
- $MPL_C = 2$ and $MPL_W = 4$
- Labor $= 25$ workers

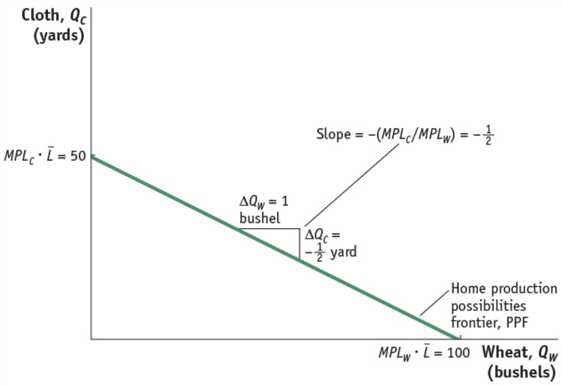

Figure 2.1
Feenstra/Taylor, *International Economics*, 4e, © 2017 by Worth Publishers

Home Country: Supply Side

- Recall that MPL is constant in each sector, so the slope of the PPF is constant

$$\text{Slope of PPF} = -\frac{MPL_C \cdot L}{MPL_W \cdot L} = -\frac{MPL_C}{MPL_W} = -\frac{1}{2}$$

- This is the **opportunity cost** of Wheat (the good on the horizontal axis)
- Suppose $Q_W \uparrow 1$, it takes one worker to produce 4 bushels of Wheat, so $1/4$ of a worker's time must now be taken away from the cloth industry.
- Since one worker can produce 2 units of cloth, $1/4$ of a worker can produce $\frac{2}{4} = \frac{1}{2}$ units of cloth.

Home Country: Demand Side

- What will actually be produced will depend on what consumers actually want; i.e. their utility function.
- For simplicity, think of a representative consumer.
- Represent their utility with indifference curves.
- What is the slope of an indifference curve?

$$\text{Slope} = -MRS = -\frac{MU_W}{MU_C}$$

Home Country: Equilibrium

- Various ways to think about the equilibrium.
- First: Graphically/intuitively
- The PPF is the country's "budget constraint" – try to get on the highest indifference curve given your budget constraint.

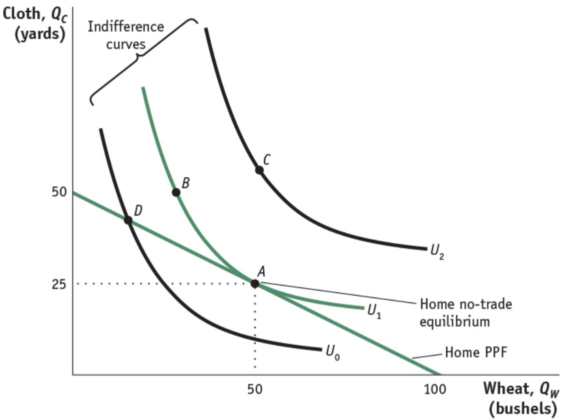

Figure 2.2
Feenstra/Taylor, *International Economics*, 4e, © 2017 by Worth Publishers

Home Country: Equilibrium

- Next think about firms:
- If firms are profit maximizing, what do we know has to be true for each firm in each sector?
- Cloth

$$\max_{L} p_C f_C(L_C) - w_C L_C$$

$$\Rightarrow p_C \frac{df_C}{dL} = w_C$$

$$\therefore p_C MPL_C = w_C$$

- Same for Wheat

$$p_W MPL_W = w_W$$

Home Country: Equilibrium

$$p_C MPL_C = w_C$$
$$p_W MPL_W = w_W$$

- What has to be true about wages in each sector? (recall every worker is identical)

$$p_W MPL_W = p_C MPL_C$$
$$\Rightarrow \frac{p_W}{p_C} = \frac{MPL_C}{MPL_W}$$

- $\frac{p_W}{p_C}$ is the **relative price** of Wheat in the home country.

Home Country: Equilibrium

$$\frac{p_W}{p_C} = \frac{MPL_C}{MPL_W}$$

- What has to be true about consumers maximizing their utility?

$$\frac{p_W}{p_C} = \frac{MU_W}{MU_C} = MRS$$

- Equilibrium prices is when supply equals demand:

$$\frac{MPL_C}{MPL_W} = \frac{p_W}{p_C} = \frac{MU_W}{MU_C}$$

- Given our technology the relative price of Wheat is $\frac{p_W}{p_C} = \frac{1}{2}$.
- Note that technology is pinning down relative prices, while utility is determining how much of each is produced in equilibrium.

Foreign Country: Supply Side

- Recall Home: $MPL_C = 2$ and $MPL_W = 4$.
- Foreign: $MPL_C^* = 1$, $MPL_W^* = 1$, and $L^* = 100$
- First, who has the absolute advantage in each good?
- Now, who has the comparative advantage in Cloth?
- Opportunity cost of Cloth in Foreign = 1 bushel of Wheat.
- Opportunity cost of Cloth in Home = 2 bushels of Wheat.
- Thus Foreign has lower opportunity cost and a *comparative advantage* in Cloth.
- Who has the comparative advantage in Wheat?
- Can a country have the comparative advantage in *both* goods?

Foreign Country: Supply Side

- Recall that MPL^* is constant in each sector, so the slope of the PPF is constant

$$\text{Slope of PPF} = -\frac{MPL_C^* \cdot L^*}{MPL_W^* \cdot L^*} = -\frac{MPL_C^*}{MPL_W^*} = -1$$

- This is the **opportunity cost** of Wheat (the good on the horizontal axis)
- Suppose $Q_W^* \uparrow 1$, it takes one worker to produce 1 bushel of Wheat, so one worker must now be taken away from the cloth industry.
- Since one worker can produce 1 units of cloth, one worker can produce 1 units of cloth.

Foreign Country: PPF

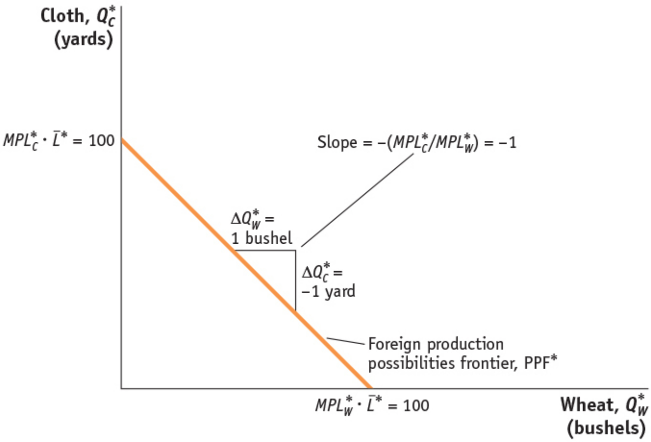

Figure 2.3
Feenstra/Taylor, *International Economics*, 4e, © 2017 by Worth Publishers

Foreign Country: Demand Side

- What will actually be produced will depend on what consumers actually want; i.e. their utility function.
- For simplicity, think of a representative consumer.
- Represent their utility with indifference curves.
- What is the slope of an indifference curve?

$$\text{Slope} = -MRS^* = -\frac{MU_W^*}{MU_C^*}$$

Foreign Country: Equilibrium

- Various ways to think about the equilibrium.
- First: Graphically/intuitively
- The PPF is the country's "budget constraint" – try to get on the highest indifference curve given your budget constraint.

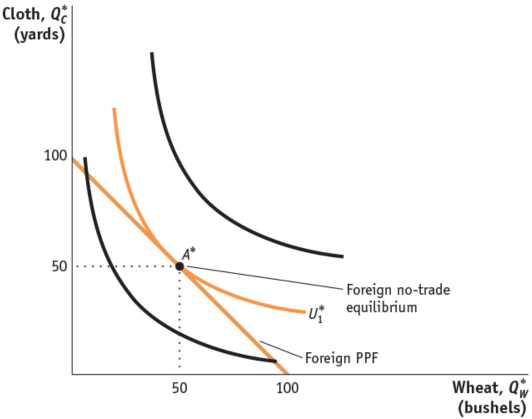

Figure 2.4

Foreign Country: Equilibrium

- Next think about firms:
- If firms are profit maximizing, what do we know has to be true for each firm in each sector?
- Cloth

$$\max_{L^*} p_C^* f_C^*(L_C) - w_C^* L_C^*$$
$$\Rightarrow p_C^* \frac{df_C^*}{dL} = w_C^*$$
$$\therefore p_C^* MPL_C^* = w_C^*$$

- Same for Wheat

$$p_W^* MPL_W^* = w_W^*$$

Foreign Country: Equilibrium

$$p_C^* MPL_C^* = w_C^*$$
$$p_W^* MPL_W^* = w_W^*$$

- What has to be true about wages in each sector? (recall every worker is identical)

$$p_W^* MPL_W^* = p_C^* MPL_C^*$$
$$\Rightarrow \frac{p_W^*}{p_C^*} = \frac{MPL_C^*}{MPL_W^*}$$

- $\frac{p_W^*}{p_C^*}$ is the **relative price** of Wheat in the foreign country.

Foreign Country: Equilibrium

$$\frac{p_W^*}{p_C^*} = \frac{MPL_C^*}{MPL_W^*}$$

- What has to be true about consumers maximizing their utility?

$$\frac{p_W^*}{p_C^*} = \frac{MU_W^*}{MU_C^*} = MRS^*$$

- Equilibrium prices is when supply equals demand:

$$\frac{MPL_C^*}{MPL_W^*} = \frac{p_W^*}{p_C^*} = \frac{MU_W^*}{MU_C^*}$$

- Given our technology the relative price of Wheat is $\frac{p_W^*}{p_C^*} = 1$.
- Note that technology is pinning down relative prices, while utility is determining how much of each is produced in equilibrium.

International Trade Equilibrium

- To begin, suppose there was a world relative price $\left(\frac{p_W}{p_C}\right)^w$ and focus on Home where the autarky equilibrium relative price of Wheat $\frac{p_W}{p_C} = \frac{1}{2}$
- For simplicity assume the nominal price $p_C = p_C^w$. What would happen if $\left(\frac{p_W}{p_C}\right)^w > \frac{p_W}{p_C} = \frac{1}{2}$?
- $p_W^W > p_W$ and from our profit maximizing wage conditions

$$p_C MPL_C = w_C \qquad \text{and} \qquad p_W^w MPL_W = w_W$$
$$\Rightarrow w_W > w_C$$

- Wheat producers get a higher price for Wheat and want to produce more Wheat.
- At the same time workers see a higher wage to work in the Wheat industry and want to move out of the Cloth industry into the Wheat industry.

International Trade Equilibrium

- Since *MPL*s are constant wages never equalize and Home FULLY specializes in Wheat.
- What about consumers?
- No home production of cloth, but can trade on the world market at $\left(\frac{p_W}{p_C}\right)^w$.
- Wheat is more expensive, but cloth is relatively cheaper, so they can exchange Wheat for more cloth than what they could have produced – **Trade Triangle**

Home Trade Triangle

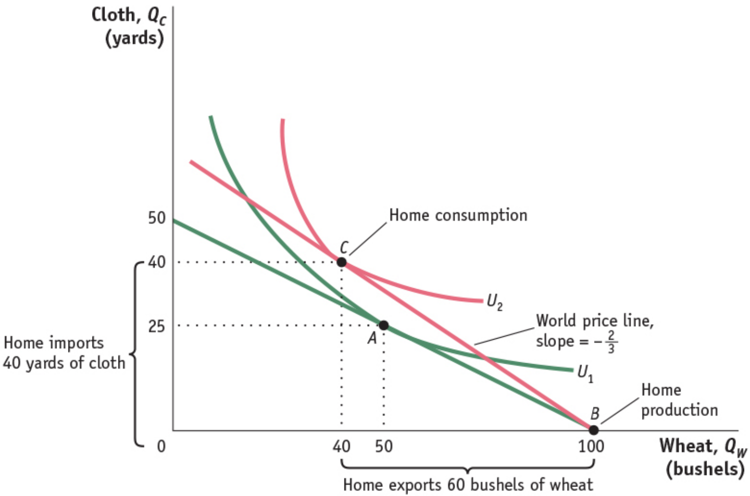

Figure 2.5

International Trade Equilibrium

- Decompose the gains from trade and specialization.
- Similar intuition for if $\left(\frac{p_W}{p_C}\right)^w < \frac{p_W}{p_C} = \frac{1}{2} \rightarrow$ Home will fully specialize in Cloth.
- Note the same thought experiment can be applied to Foreign.

Foreign Trade Triangle

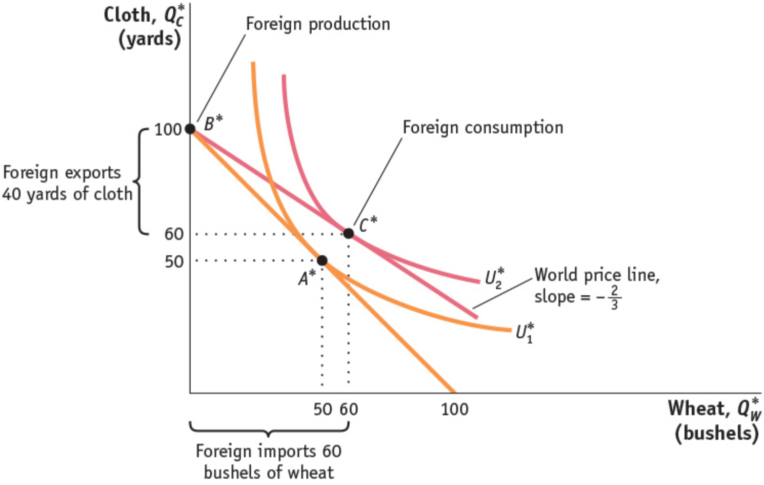

Figure 2.6
Feenstra/Taylor, *International Economics*, 4e, © 2017 by Worth Publishers

International Trade Equilibrium

- Now recall the autarky equilibrium relative prices of Wheat $\frac{p_W}{p_C} = \frac{1}{2}$ for Home and $\frac{p_W^*}{p_C^*} = 1$ for Foreign.
- Can there be a trade equilibrium between the two countries where $\left(\frac{p_W}{p_C}\right)^w > \frac{p_W^*}{p_C^*} = 1$?
- NO. Both countries would only want to produce Wheat... which is fine if consumers *only* want to be full and naked.
- Can there be a trade equilibrium between the two countries where $\left(\frac{p_W}{p_C}\right)^w < \frac{p_W}{p_C} = \frac{1}{2}$?
- NO. Both countries would only want to produce cloth... which is fine if consumers aren't hungry.
- So the equilibrium **Terms of Trade** must be

$$\frac{1}{2} = \frac{p_W}{p_C} < \left(\frac{p_W}{p_C}\right)^w < \frac{p_W^*}{p_C^*} = 1$$

International Trade Equilibrium

- Suppose $\left(\frac{p_W}{p_C}\right)^w = \frac{2}{3}$.
- We just saw how the country as a whole is better with trade when the TOT is strictly different from their autarky relative price.
- How are wages affected in each country?
- First let's think about the real wage. So we need to divide the wage by a price

International Trade Equilibrium

- First in terms of Cloth
- For Home

$$\text{Autarky} \quad \frac{w}{p_C} = \left(\frac{p_W}{p_C}\right) MPL_W = \left(\frac{1}{2}\right) 4 = 2 \text{ yards}$$

$$\text{Trade} \quad \frac{w}{p_C^w} = \left(\frac{p_W}{p_C}\right)^w MPL_W = \left(\frac{2}{3}\right) 4 = \frac{8}{3} > 2 \text{ yards}$$

- For Foreign

$$\text{Autarky} \qquad \frac{w^*}{p_C^*} = MPL_C^* = 1 \text{ yard}$$

$$\text{Trade} \qquad \frac{w^*}{p_C^w} = MPL_C^* = 1 \text{ yard}$$

International Trade Equilibrium

- Now in terms of Wheat
- For Home

$$\begin{aligned} \text{Autarky} \qquad & \frac{w}{p_W} = MPL_W = 4 \text{ bushels} \\ \text{Trade} \qquad & \frac{w}{p_W^w} = MPL_W = 4 \text{ bushels} \end{aligned}$$

- For Foreign

$$\begin{aligned} \text{Autarky} \quad & \frac{w^*}{p_W^*} = \left(\frac{p_C^*}{p_W^*}\right) MPL_C^* = 1 \text{ bushel} \\ \text{Trade} \quad & \frac{w^*}{p_W^w} = \left(\frac{p_C}{p_W}\right)^w MPL_C^* = \left(\frac{3}{2}\right) 1 = \frac{3}{2} > 1 \text{ bushel} \end{aligned}$$

International Trade Equilibrium

- The real wage in terms of the good each country has a comparative advantage in stays the same, but the real wage in terms of the other good goes up.
- In real terms, both wages increase.
- However the real wage of Home is higher than that of Foreign.
- This is driven by absolute advantage. In the end, wages are driven by productivity and Home is more productive than Foreign.

International Trade Equilibrium

- Before, we arbitrarily chose a world price (Terms of Trade). How do we determine what this actually will be?
- Need to incorporate demand.
- First, let's derive the Home's export supply of Wheat.
- For a given relative price p_W/p_C home will be willing to supply certain bundles of Wheat and Cloth.
- However, there will be only one specific bundle demanded at that price – recall the Trade Triangle.
- Now, try various different relative prices.

Home Export Supply

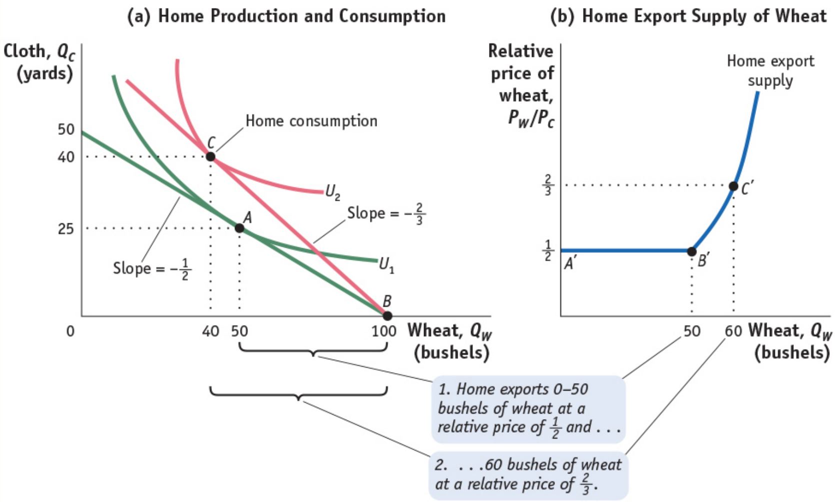

Figure 2.9
Feenstra/Taylor, *International Economics*, 4e, © 2017 by Worth Publishers

International Trade Equilibrium

- Now, do the same thought experiment with Foreign demand for wheat.

Foreign Import Demand

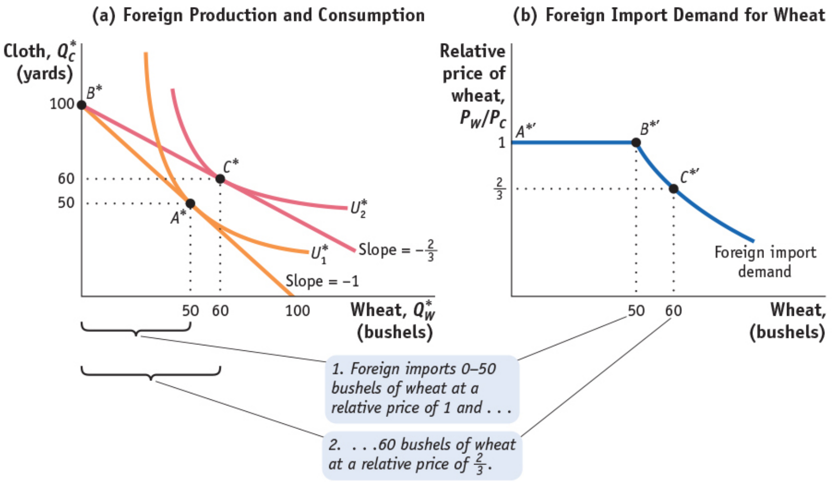

Figure 2.10
Feenstra/Taylor, *International Economics*, 4e, © 2017 by Worth Publishers

Putting Supply and Demand Together

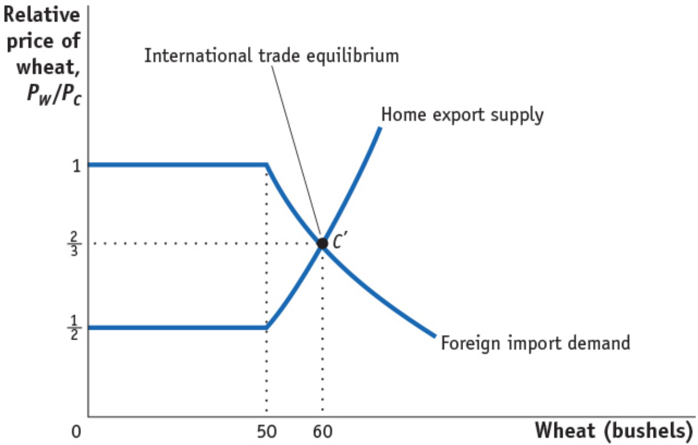

Figure 2.11
Feenstra/Taylor, *International Economics*, 4e, © 2017 by Worth Publishers

Thought Experiment

- What if Home got really big?
- Then their export supply would shift out.
- If it got so big the relative price of home would not change from autarky.
- **Implication:** Small countries gain the most from free trade, but no country can be made *worse off*.

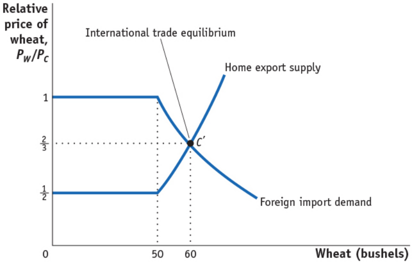

Figure 2.11
Feenstra/Taylor, *International Economics*, 4e, © 2017 by Worth Publishers

Size Does Matter

- **Importance of being unimportant:** When small countries trade with big countries, the small are likely to enjoy most of the mutual gains from trade.
- **However**, in reality, his isn't always the case, note McLaren (1997) and Cole, Haley, and Lowen (2008).

Criticisms of the Ricardian Model

- Everyone gains without any sort of redistribution.
- Too much specialization.
- Labor only input.
- Exogenous technology. Why doesn't information about production techniques transfer across countries??
- Not a ton of empirical support.

- Next up "Gains and Losses from Trade in the Specific-Factors Model"